

C++20 MESH REDUCTION · GAME-ENGINE IMPORT PIPELINES

Fewer triangles. Measured transitions.

10 actual LOD chains, including 6 new nature examples. Inspect every level, its triangle count, measured bake time and vertex storage mode.

Geometry, with an error budget.

Propose a simpler mesh. Check it against the previous level and the original. Keep the smallest valid chain found.

Rows 01–04 and the score charts are archived protocol v1 results. The six nature chains use the current library with geometry-only coverage checks and have no SCORE. Opacity textures and shading are outside those checks.

FROZEN MESH CORPUS 120 30 rocks · 30 organics 40 manufactured · 20 complex scans	512 → 16 px / COVERAGE PILOT 80.18 SCORE +4.31 points · same eight assets and visual gates	FROZEN VALIDATION / 20 MESHES 86.48 SCORE +6.18 points · 20/20 completed · 0 failures	INTEGRATION PATHS 3 chain modes Direct · progressive · hybrid Rebuild vertices or reuse source buffers
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01 / ACTUAL OUTPUT GEOMETRY

From source to 16 pixels.

Each tile is an actual exported LOD mesh. Nature rows audit full card geometry; texture opacity is not scored.

DISPLAY	ClayWireframeSilhouette	LEVELS	Audit slotsRuntime levels	SCALE	EnlargedTarget pxReset views ↗
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Enlarged for inspection. Same camera and source scale across each row. Drag a mesh to rotate its chain; click a tile to inspect it.

01 / ROCK

Moon rock

Rock surface detail gives ways to the permitted distant silhouette. · 8 runtime levels / 8 audit slots

Bake pipeline: 48.80 s · Vertices: rebuilt / new buffers · time includes import, audit and export

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

3,304 tris

UNCHANGED · SOURCE

SOURCE VERTICES

2,974 tris

✓ Δ 1.41 / 2.00

NEW VERTICES

2,676 tris

✓ Δ 1.96 / 2.17

NEW VERTICES

268 tris

✓ Δ 2.31 / 2.33

NEW VERTICES

134 tris

✓ Δ 2.47 / 2.50

NEW VERTICES

68 tris

✓ Δ 2.16 / 2.67

NEW VERTICES

34 tris

✓ Δ 2.46 / 2.83

NEW VERTICES

18 tris

✓ Δ 2.16 / 3.00

NEW VERTICES

−99.55% at LOD 7

3,304 → 18 tris

02 / ORGANIC

Quiver tree trunk

The extreme tail passes pixel-distance gates but loses filled area. · 8 runtime levels / 8 audit slots

Bake pipeline: 75.89 s · Vertices: rebuilt / new buffers · time includes import, audit and export

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

17,978 tris

UNCHANGED · SOURCE

SOURCE VERTICES

1,066 tris

✓ Δ 1.98 / 2.00

NEW VERTICES

534 tris

✓ Δ 2.05 / 2.17

NEW VERTICES

268 tris

✓ Δ 2.29 / 2.33

NEW VERTICES

134 tris

✓ Δ 2.12 / 2.50

NEW VERTICES

68 tris

✓ Δ 2.51 / 2.67

NEW VERTICES

6 tris

✓ Δ 2.72 / 2.83

NEW VERTICES

2 tris

✓ Δ 1.46 / 3.00

NEW VERTICES

−99.99% at LOD 7

17,978 → 2 tris

03 / MANUFACTURED

Metal stool

This asset still needs better intermediate proposals. · 6 runtime levels / 8 audit slots

Bake pipeline: 45.19 s · Vertices: rebuilt / new buffers · time includes import, audit and export

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

6,532 tris

UNCHANGED · SOURCE

SOURCE VERTICES

1,981 tris

✓ Δ 1.71 / 2.00

NEW VERTICES

990 tris

✓ Δ 1.96 / 2.17

NEW VERTICES

495 tris

✓ Δ 1.77 / 2.33

NEW VERTICES

247 tris

✓ Δ 2.47 / 2.50

NEW VERTICES

247 tris

✓ Δ 0.00 / 2.67

NEW VERTICES

124 tris

✓ Δ 2.16 / 2.83

NEW VERTICES

124 tris

✓ Δ 0.00 / 3.00

NEW VERTICES

−98.1% at LOD 7

6,532 → 124 tris

04 / COMPLEX SCAN

Bell X-1

Repeated audit slots can share one runtime mesh. · 5 runtime levels / 8 audit slots

Bake pipeline: 57.71 s · Vertices: rebuilt / new buffers · time includes import, audit and export

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

99,586 tris

UNCHANGED · SOURCE

SOURCE VERTICES

10,180 tris

✓ Δ 1.46 / 2.00

NEW VERTICES

10,180 tris

✓ Δ 0.00 / 2.17

NEW VERTICES

5,090 tris

✓ Δ 2.05 / 2.33

NEW VERTICES

2,546 tris

✓ Δ 1.71 / 2.50

NEW VERTICES

2,546 tris

✓ Δ 0.00 / 2.67

NEW VERTICES

1,273 tris

✓ Δ 2.83 / 2.83

NEW VERTICES

1,273 tris

✓ Δ 0.00 / 3.00

NEW VERTICES

−98.7% at LOD 7

99,586 → 1,273 tris

05 / NATURE · GRASS

Grass clump

Card geometry coverage; texture opacity and shading are not audited. · 4 runtime levels / 8 audit slots

Chain bake: 2.22 s · Vertices: rebuilt / new buffers · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

2,489 tris

UNCHANGED · SOURCE

SOURCE VERTICES

2,489 tris

✓ Δ 0.00 / 2.00

SOURCE VERTICES

2,489 tris

✓ Δ 0.00 / 2.17

SOURCE VERTICES

2,240 tris

✓ Δ 2.05 / 2.33

NEW VERTICES

2,016 tris

✓ Δ 1.71 / 2.50

NEW VERTICES

2,016 tris

✓ Δ 0.00 / 2.67

NEW VERTICES

1,008 tris

✓ Δ 2.72 / 2.83

NEW VERTICES

1,008 tris

✓ Δ 0.00 / 3.00

NEW VERTICES

−59.5% at LOD 7

2,489 → 1,008 tris

06 / NATURE · FERN

Fern fronds

Card geometry coverage; texture opacity and shading are not audited. · 5 runtime levels / 8 audit slots

Chain bake: 3.61 s · Vertices: shared original buffer · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

2,384 tris

UNCHANGED · SOURCE

SOURCE VERTICES

2,384 tris

✓ Δ 0.00 / 2.00

SOURCE VERTICES

2,384 tris

✓ Δ 0.00 / 2.17

SOURCE VERTICES

2,145 tris

✓ Δ 2.31 / 2.33

SOURCE VERTICES

1,930 tris

✓ Δ 1.50 / 2.50

SOURCE VERTICES

965 tris

✓ Δ 2.31 / 2.67

SOURCE VERTICES

482 tris

✓ Δ 1.71 / 2.83

SOURCE VERTICES

482 tris

✓ Δ 0.00 / 3.00

SOURCE VERTICES

−79.8% at LOD 7

2,384 → 482 tris

07 / NATURE · SHRUB

Leafy shrub

Card geometry coverage; texture opacity and shading are not audited. · 7 runtime levels / 8 audit slots

Chain bake: 6.10 s · Vertices: rebuilt / new buffers · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

7,590 tris

UNCHANGED · SOURCE

SOURCE VERTICES

7,590 tris

✓ Δ 0.00 / 2.00

SOURCE VERTICES

6,831 tris

✓ Δ 1.96 / 2.17

NEW VERTICES

6,147 tris

✓ Δ 1.46 / 2.33

NEW VERTICES

3,073 tris

✓ Δ 1.61 / 2.50

NEW VERTICES

1,536 tris

✓ Δ 2.61 / 2.67

NEW VERTICES

768 tris

✓ Δ 2.82 / 2.83

NEW VERTICES

94 tris

✓ Δ 2.46 / 3.00

NEW VERTICES

−98.8% at LOD 7

7,590 → 94 tris

08 / NATURE · TREE

Broadleaf tree

Card geometry coverage; texture opacity and shading are not audited. · 3 runtime levels / 8 audit slots

Chain bake: 0.70 s · Vertices: rebuilt / new buffers · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

870 tris

UNCHANGED · SOURCE

SOURCE VERTICES

870 tris

✓ Δ 0.00 / 2.00

SOURCE VERTICES

870 tris

✓ Δ 0.00 / 2.17

SOURCE VERTICES

870 tris

✓ Δ 0.00 / 2.33

SOURCE VERTICES

870 tris

✓ Δ 0.00 / 2.50

SOURCE VERTICES

870 tris

✓ Δ 0.00 / 2.67

SOURCE VERTICES

783 tris

✓ Δ 2.31 / 2.83

NEW VERTICES

704 tris

✓ Δ 0.96 / 3.00

NEW VERTICES

−19.1% at LOD 7

870 → 704 tris

09 / NATURE · FLOWERS

Flowering ground cover

Card geometry coverage; texture opacity and shading are not audited. · 6 runtime levels / 8 audit slots

Chain bake: 13.51 s · Vertices: rebuilt / new buffers · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

17,593 tris

UNCHANGED · SOURCE

SOURCE VERTICES

15,833 tris

✓ Δ 1.74 / 2.00

NEW VERTICES

15,833 tris

✓ Δ 0.00 / 2.17

NEW VERTICES

14,249 tris

✓ Δ 2.29 / 2.33

NEW VERTICES

14,249 tris

✓ Δ 0.00 / 2.50

NEW VERTICES

7,124 tris

✓ Δ 2.21 / 2.67

NEW VERTICES

3,562 tris

✓ Δ 1.50 / 2.83

NEW VERTICES

356 tris

✓ Δ 2.98 / 3.00

NEW VERTICES

−98.0% at LOD 7

17,593 → 356 tris

10 / NATURE · BRANCH

Leafy branch

Card geometry coverage; texture opacity and shading are not audited. · 3 runtime levels / 8 audit slots

Chain bake: 0.92 s · Vertices: rebuilt / new buffers · generation + audit

LOD 0512.0 px

LOD 1312.1 px

LOD 2190.2 px

LOD 3115.9 px

LOD 470.7 px

LOD 543.1 px

LOD 626.3 px

LOD 716.0 px

120 tris

UNCHANGED · SOURCE

SOURCE VERTICES

120 tris

✓ Δ 0.00 / 2.00

SOURCE VERTICES

120 tris

✓ Δ 0.00 / 2.17

SOURCE VERTICES

120 tris

✓ Δ 0.00 / 2.33

SOURCE VERTICES

120 tris

✓ Δ 0.00 / 2.50

SOURCE VERTICES

62 tris

✓ Δ 2.66 / 2.67

NEW VERTICES

62 tris

✓ Δ 0.00 / 2.83

NEW VERTICES

31 tris

✓ Δ 2.92 / 3.00

NEW VERTICES

−74.2% at LOD 7

120 → 31 tris

Reading the tiles: Δ is audited upper error against the previous LOD, followed by its pixel limit. Each reduced mesh also passed its direct-to-source limit. LOD 0 is unchanged.

Neutral clay uses geometric face normals, without textures. Presentation angles are not audit cameras. A low coverage error does not constrain shading.

02 / COMPARABLE COVERAGE SCORES Useful final LODs need better search. Preventing new topology locks helps coarse reductions. Avoiding duplicate hybrid inputs and using the full existing proposal budget improves the complete chain score. Previous implementation Relax topology locks Prevent new locks Guard + better search Correct input ownership 75.86 75.68 75.75 80.18 80.18 050100 Same eight development assets, 512 → 16 px, hybrid rebuild, source cap 8 px, eight proposals per level, unchanged cameras and gates. SCORE = 100 × (1 − category-balanced mean retained triangle ratio). Exact duplicate compaction does not change SCORE.	03 / WHAT THE EXPERIMENTS SAY Smaller tails, with measured limits. −48.9% FINAL RETAINED RATIO 53/64 RUNTIME / AUDIT SLOTS 2 vs 34 TREE / CAP 8 vs 2 The last levels use fewer triangles. Category-balanced final retained triangles fall from 3.14% to 1.61%. Last-three-level retained ratio falls 50.9%. Gates and SCORE are unchanged. Exact repeats share one mesh. The C/C++ API and glTF export expose compact runtime levels. Every scheduled source and transition audit stays in the manifest and in the score. A loose distance budget can lose filled area. The 2-triangle tail passes the denser 706-camera rectangle but loses much of the thin trunk's coverage. Its smaller quad cost includes that coverage loss. Shading is outside this coverage-only run.
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